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RESEARCH ARTICLE

A Hybrid CNN-LSTM Model With Attention Mechanism for Improved Intrusion Detection in Wireless IoT Sensor Networks

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ABSTRACT Wireless Internet of Things (IoT) Sensor Networks (WIoTSNs) are frequently deployed in resource-constrained environments where security threats pose significant challenges. Existing intrusion detection systems (c) often struggle with scalability and efficiency under the unique demands of IoT networks. This work introduces an Intrusion Detection System (IDS) framework that integrates Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks in a hybrid architecture, enhanced by an attention mechanism to improve feature extraction and classification accuracy. To address computational demands, an enhanced Particle Swarm Optimization (PSO) algorithm is implemented for dynamic feature selection, thereby optimizing the system's efficiency in high-dimensional data environments characteristic of IoT networks. The proposed model enhances IoT intrusion detection by integrating a novel hybrid CNN-LSTM with an attention mechanism, thereby improving feature extraction and temporal pattern recognition. Additionally, the improved dynamic PSO algorithm optimizes feature selection in real time, enhancing classification accuracy and adaptability to evolving IoT network threats. This combination ensures more efficient and robust intrusion detection in dynamic IoT environments. Experimental evaluations using a standard IoT intrusion dataset indicate that the proposed model achieves notable accuracy rates of 98.73% with CNN, 99.87% with LSTM, 99.12% with CNN-LSTM, and 98.88% with the enhanced CNN-LSTM with attention, demonstrating an improvement over existing techniques. The framework's resilience and adaptability underscore its potential for enhancing network security in real-world IoT applications by addressing evolving threats and computational constraints.

INDEX TERMS CNN, LSTM, IDS, attention mechanism, wireless Internet of Things sensor networks (WIoTSNs).

I. INTRODUCTION

Wireless IoT Sensor Networks (WIoTSNs) stand at the forefront of contemporary technological progress, revolutionising data acquisition, monitoring, and control. Their pervasive application across various domains indicates a move toward smart, interconnected systems. The increasing integration of WIoTSNs in critical systems underscores their transformative impact on industrial processes, smart cities, healthcare, agriculture, environmental monitoring, and energy management.

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Their versatility addresses current challenges and contributes to the development of a more interconnected and efficient global landscape. As WIoTSNs become increasingly prevalent, addressing security challenges, particularly in implementing robust intrusion detection mechanisms, becomes crucial to ensure the resilience and dependability of vital systems across diverse sectors [1]. IoT networks encounter significant security issues stemming from the extensive array of connected devices, many of which lack robust security measures. This vulnerability makes them prone to unauthorized access and potential data compromises. Because these networks are decentralized, there is a greater chance of eavesdropping and unwanted access. Communication security is ensured through various methods, including access control, optimization-driven secure routing techniques, agent-based approaches, temporal analysis, and intrusion detection methods designed for feature selection and classification [2].

Wireless IoT Sensor Networks (WIoTSNs) are revolutionizing data acquisition and monitoring across various domains, including smart cities, healthcare, and energy management. However, the decentralized nature and resource constraints of these networks render them susceptible to security threats. Recent high-quality studies, such as [3], have explored throughput maximization and AoI constraints in IoT environments, underscoring the need for robust security solutions. The proposed study builds on these advancements by addressing security challenges in IoT environments using an enhanced hybrid CNN-LSTM model. This model integrates dynamic PSO for feature selection, optimizing detection accuracy in resource-constrained settings. Unlike conventional IDS methods, which often struggle with scalability and adaptability, the proposed framework utilizes deep learning and metaheuristic optimization to achieve superior performance.

Intrusion detection is crucial for identifying and mitigating security threats, especially in dynamic IoT environments. Robust mechanisms, incorporating anomaly detection and machine learning, are essential for adapting to IoT's diverse and expanding networks [4], [5], [6], [7]. As IoT networks grow, secure communication protocols and encryption are vital for data protection. However, traditional intrusion detection systems face challenges in scaling and adapting to IoT's unique complexities, such as resource constraints and evolving cyber threats [8], [9], [10] There is a pressing need for advanced intrusion detection tailored to the intricacies of IoT, ensuring timely threat identification and the resilience of connected systems.

Over the past decade, IDSs have increasingly incorporated machine learning (ML) or deep neural network (DNN) techniques to enhance their ability to detect attacks. Deep learning (DL), a subset of ML, which employs extensive DNN networks with multiple hidden layers to address complex challenges across various applications [11] Deep learning, a specialized branch of artificial neural networks (ANNs) in machine learning, is distinguished by its use of numerous hidden layers. It has been successfully applied in various research fields, including speech recognition, image processing, and natural language processing. Deep learning can interpret large, multidimensional datasets, perform feature selection, and classify and identify correlations within the data [12]. The input for each layer is generated directly from the output of the layer preceding it. While DL achieves high performance, its primary drawback lies in the extended training time required, especially with larger datasets.

CNNs and RNNs are among the most popular types of deep learning models. CNNs are specialized feed-forward architectures designed to handle input data organized in array formats, commonly applied in image-processing tasks [13]. CNNs are designed to resemble the brain, which contains a specific region known as the receptive field, where each neuron responds to stimuli within its respective field, collectively forming the entire visual area. The core architecture of CNN consists of three layers: the convolution layer, the pooling layer, and the classification layer [14].

Recursive Neural Networks (RNNs) excel in handling sequential data, making them indispensable for immediate intrusion detection and anomaly identification, which are vital for protecting IoT networks from unauthorized access and security breaches. Conversely, DRNNs extend this capability by incorporating dynamic mechanisms, allowing for more flexible and adaptive modeling of sequential data [15]. This adaptability can be particularly advantageous in IoT environments where data patterns may evolve or exhibit dynamic behavior. Consequently, while RNNs offer robust sequential data processing capabilities, DRNNs provide enhanced flexibility and adaptability, potentially yielding superior performance in IoT data security applications.

A type of RNN designed primarily to maintain and manage long-term dependencies is called an LSTM network. It overcomes the vanishing gradient issue found in traditional RNNs by using gated cells to store information. LSTM functions like computer memory, allowing it to read, write, and retain information. LSTM networks use three separate gates to address the vanishing gradient problem: the forget gate determines what information stays in the cell, the input gate controls the addition of new information, and the output gate keeps information that might be useful for subsequent stages [16].

The attention mechanism significantly enhances the efficiency and accuracy of processing perceptual information. It serves as a resource allocation strategy to address information overload by prioritizing important data with limited computing resources [17]. Consequently, researchers have begun applying attention mechanisms in computer vision [18]; for example, a saliency-based model was proposed to extract local low-level visual features for identifying salient regions. In the neural network domain, attention mechanisms have been utilized in recurrent neural networks for

image classification [19], and by [20] This is for simultaneous translation and alignment in machine translation tasks. The attention mechanism has emerged as a crucial element in numerous neural architectures, with applications in image captioning, text classification, machine translation, action recognition, and image analysis.

This paper presents an advanced hybrid approach that integrates metaheuristic algorithms with deep learning techniques to address security challenges on the Internet of Things (IoT). Specifically, it combines a modified metaheuristic algorithm with CNN and LSTM networks to enhance attack detection in IoT environments. This study focuses on developing a robust intrusion detection system tailored to the dynamic and resource-limited characteristics of wireless IoT sensor networks. Utilizing a hybrid CNN-LSTM model to enhance feature extraction and improve attack classification, the research evaluates the system's performance using standard statistical metrics and benchmarks against advanced neural network techniques. This work offers practical solutions to enhance adaptability and effectiveness in IoT sensor networks, providing a robust framework for securing IoT environments. The research contribution outlined in this paper is as follows:

- Examine current research on intrusion detection methods for wireless Internet of Things sensor networks, with an emphasis on applying artificial neural networks to WIoTSN intrusion detection.
- Develop a hybrid CNN-LSTM model to improve feature extraction for attack classification in an intrusion detection model for Internet of Things environments.
- Assessment using Standard statistical measures to evaluate the performance of the suggested intrusion detection mechanism.
- Evaluation of the proposed method in relation to advanced neural network techniques for feature extraction and attack classification.

The remainder of the document is organized as follows: Section II reviews relevant works. Section III discusses the methodology of the proposed model. Section IV covers the study's findings. Section V provides a summary of the paper and discusses potential avenues for future research, providing valuable insights for researchers regarding areas that warrant further exploration in federated learning-enhanced intrusion detection.

II. RELATED WORKS

As IoT continues to expand rapidly, Wireless IoT Sensor Networks (WIoTSN) have emerged as prime targets for cyberattacks, largely due to the widespread deployment of sensor devices that gather and transmit data to the cloud. These interconnected devices in intelligent infrastructures, including smart homes and buildings, create vast attack surfaces for cyber intrusions. Several IDS have been developed to tackle these challenges, utilizing artificial intelligence and deep learning to improve security. IDS are critical for securing IoT environments against cyber threats. Deep Learning (DL) encompasses elements for feature extraction as well as deep neural network architectures. DL automatically extract features from large datasets, eliminating the need for manual feature engineering and exhibiting robust capabilities in multi-class classification. As a result, DL is rapidly becoming a key area of research in IoT intrusion detection.

In the research conducted by [21], the authors introduced a technique aimed at enhancing the detection of DoS attacks. In IoT networks, they have launched distributed denial-of-service (DDoS) attacks by integrating artificial intelligence-based technology, particularly deep learning models. By utilizing a CNN model for analyzing network data and training the RNN state model, the proposed approach significantly improved accuracy for DoS and DDoS detection. Through dual deployment, the method achieved an impressive 99.99% accuracy in detecting DoS and DDoS attacks, as well as a high accuracy of 98% in identifying various attack patterns. These results outperformed those of the latest technologies, underscoring the approach's efficiency in enhancing the security of IoT networks to protect against advanced cyber threats.

Authors in [22] presented an innovative approach for detecting IoT network attacks using a dual CNN-CNN strategy. The first CNN model was employed for feature selection, identifying key characteristics from raw network traffic data that are relevant for attack detection. The second CNN model then used these identified features to develop a reliable detection system. When assessed with the BoT-IoT 2020 dataset, the proposed approach demonstrated excellent performance, achieving a detection accuracy of 98.04%, precision of 98.09%, recall of 99.85%, and a low false positive rate (FPR) of 1.93%. This approach surpassed conventional feature selection and detection methods by independently learning hierarchical features from the data, thereby improving both detection accuracy and efficiency. While the approach demonstrated significant strengths in attack detection, future work may focus on enhancing detection capabilities by incorporating more complex architectures, exploring additional feature selection methods, and developing rigorous mathematical models to further validate the approach's effectiveness.

The work presented by [23] addressed security challenges in IoT networks by proposing a novel training algorithm for DL-based IDS. The authors introduced an innovative neighborhood search method using a particle swarm optimization algorithm (NSBPSO). The method was designed to improve the capabilities of conventional PSO methods by integrating concepts such as employed bees, onlooker bees, and multi-parent crossover. The enhanced NSBPSO algorithm was subsequently employed to effectively train a deep convolutional neural network (DCNN) for intrusion detection purposes. The NSBPSO-based DCNN demonstrated improved accuracy and performance on the UNSW-NB15 and Bot-IoT datasets compared to existing intrusion detection system (IDS) techniques. While the approach enhanced

detection capabilities and parameter tuning, the work could be improved by reducing computational costs and improving detection rates.

Reference [24] presented a novel IDS designed to identify DDoS attacks in IoT networks through the integration of DL and a multi-objective optimization approach. The IDS utilized a Jumping Gene-adapted NSGA-II algorithm for data dimensionality reduction, followed by a hybrid CNN integrated with LSTM for classifying attacks. Extensive experiments were conducted using the CICIDS2017 dataset on a high-performance computer (HPC) with GPU support. The results demonstrated a 99.03% accuracy, a 99.36% F1-score, and a 5-fold decrease in training duration, outperforming state-of-the-art methods. Despite its high accuracy and reduced computational cost, the system's deployment in real-time distributed IoT environments could face scalability and complexity challenges.

The authors in [25] proposed a hybrid CNN-LSTM model for intrusion detection within IoT environments, combining CNN and LSTM architectures and a hybrid CNN-LSTM framework that utilized PSO for feature selection and dimensionality reduction. While this approach achieved promising results, particularly with LSTM reaching 99.82% accuracy and significantly enhancing IoT security the model faced several limitations. The static nature of PSO did not adapt to evolving IoT data, potentially including irrelevant features and excluding important new ones. Furthermore, the model's lack of an attention mechanism diminished its capacity to focus on important patterns, thereby increasing the risk of false detections. The framework also lacked dynamic adaptability to real-time data changes, and the high dimensionality of IoT data posed a risk of overfitting.

The work presented in [26] introduced a Self-Attention-based Deep Convolutional Neural Network (SA-DCNN) model, which improved feature extraction and anomaly detection by leveraging self-attention mechanisms. This approach effectively prioritized critical features in network traffic data, thereby enhancing classification performance and reducing the number of false positives. The integration of self-attention mechanisms refined spatial dependencies in the data, allowing the model to focus on the most relevant attack patterns, which led to superior detection accuracy.

Ensemble learning techniques have emerged as a powerful approach for improving intrusion detection systems (IDS) in IoT environments by enhancing model interpretability and detection accuracy. Traditional IDS methods often struggle with high-dimensional data and evolving attack patterns, necessitating more adaptable solutions. The study in [27] explored the effectiveness of ensemble models, particularly blending techniques, in classifying cyber threats while providing interpretable insights into IDS decision-making. By integrating multiple models and leveraging blending techniques, the strengths of diverse classifiers were effectively combined, thereby improving detection precision and reducing false positives. Additionally, the study highlighted the importance of explainable AI methods, such as Local Interpretable Model-Agnostic Explanations (LIME) and counterfactual explanations, to enhance IDS transparency and usability. These findings reinforced the growing relevance of ensemble-based approaches in securing IoT networks against sophisticated cyber threats.

Existing works demonstrate several strengths in leveraging advanced DL techniques for the detection of intrusions in IoT-based wireless sensor networks. Using CNNs, LSTMs, and hybrid models has enhanced accuracy and efficiency in detecting complex attacks, such as DDoS, DoS, and other threats. Techniques such as CNN for feature selection and hybrid CNN-LSTM models integrated with multi-objective optimization (e.g., NSGA-II) and Particle Swarm Optimization (PSO) have achieved impressive detection accuracies, ranging from 98% to 99.99%, along with reduced false positive rates and computational costs. Additionally, innovations such as dynamic feature selection and attention mechanisms promise to enhance detection precision and facilitate real-time adaptability. However, several limitations remained, including challenges with scalability, especially in distributed IoT environments, the static nature of some optimization algorithms (e.g., PSO), which struggle with evolving data, and the risk of overfitting due to the high dimensionality of IoT data. Moreover, models lacking attention mechanisms may miss critical patterns, and the computational complexity of some frameworks poses issues for real-world deployment. This work aims to address these challenges by introducing a novel algorithm that offers enhanced adaptability, dynamic feature selection, and improved real-world performance in IoT intrusion detection.

Despite these efforts, deep learning-based IDS for IoT-based networks remain underexplored, partly due to the lack of comprehensive datasets that include attack scenarios. This research aims to bridge this gap by advancing deep learning methodologies tailored for IoT security.

III. METHODOLOGY

This study's methodology involves creating and assessing four deep learning models for identifying network intrusions in Internet of Things networks: a CNN, an LSTM, a hybrid CNN-LSTM model, and an improved CNN-LSTM model. The dataset used for this analysis was preprocessed and subjected to dimensionality reduction to enhance performance and reduce computational complexity.

List of Notations:

Table 1 summarizes all key notations used throughout the paper.

A. DATASET

An IoT Network Intrusion Dataset provided, contained the necessary data to train and evaluate the neural network model, specifically designed to identify potential security breaches within IoT environments. This dataset comprised 20 carefully selected features that represent various aspects of network

TABLE 1. List of notations.

Symbol	Description
v	Velocity of a particle in PSO
x	Position of a particle in PSO
w	Inertia weight
$c_1 c_2$	Acceleration coefficients for PSO
$r_1 r_2$	Random values for stochastic updates in PSO
$\sum$	Summation operation
$*$	Convolution operation in CNN
h	Hidden state in LSTM
f	Activation function
σ	Standard deviation
μ	Mean value
t	Time step in sequence-based models
b	Bias term in neural networks
R	Pearson correlation coefficient

traffic. These features include attributes such as 'Src_IP', 'Flow_Pkts/s' and, 'Fwd_Pkt_Len_Min' which capture critical information about the source of the traffic, the packet flow rate, and the minimum length of forward packets, respectively. The target variable in this dataset was a binary label indicating whether the observed traffic was normal or represented an anomaly, making it suitable for binary classification tasks. The selection of these features was designed to optimize the model's ability to detect minor variations in network behavior, thereby enhancing the overall accuracy of intrusion detection in IoT networks. Table 2 lists the different types of attacks in the IoTID20 dataset, while Figure 1 presents the number of features associated with each class label.

B. PARTICLE SWARM OPTIMIZATION (PSO) METHOD

The PSO method is proposed for recognizing essential features which help in identifying malicious attacks in network datasets [28], [29]. The technique was introduced by Eberhart and Kennedy and is recognized as an effective and widely respected global search method. The PSO algorithm is regarded as a practical solution due to its straightforward feature coding, global search capabilities, computational efficiency, minimal parameter requirements, and ease of implementation for addressing and selecting important features [30]. PSO is employed to identify significant features.

Algorithm 1 illustrates the procedure employed to select essential features from an intrusion detection network dataset. PSO employs a principal space approach to examine the search space, using a subset of key components selected as features. In this method, particles represent potential solutions within the population, collectively referred to as a swarm, as they navigate the search space. Particles are generated by randomly assigning 1s and 0s, where a principal component with a value of 1 is selected, while a component with a value of 0 is ignored. To enhance PSO's effectiveness, it operates randomly, exploring the search space to find the optimal feature subset by continuously updating the positions and velocities of the particles [31], [32].

The PSO method was chosen for feature selection due to its simplicity, computational efficiency, and suitability for high-dimensional IoT data. Compared to Genetic Algorithms (GAs) and Ant Colony Optimization (ACOs), PSO achieves faster convergence and requires fewer computational resources, making it ideal for resource-constrained Internet of Things (IoT) environments. By dynamically adjusting feature subsets in response to real-time data, the PSO algorithm enhances detection accuracy and adaptability. The position and velocity of particle *the following equations represent me*:

$$v_{id}^{t+1} = wv_{id}^{t} + c_1 r_{1i}\left(p_{id} - x_{id}^{t}\right) + c_2 r_{2i}\left(g_{\mathrm{gd}} - x_{id}^{t}\right), \quad (1)$$

$$x_i^{t+1} = x_i^{t} + v_i^{t+1}, \quad (2)$$

where:

- v_i^t : Velocity of particle i at iteration t
- x_i^t : Position of particle i at iteration t.
- w : Inertia weight, balancing exploration and exploitation.
- $c_1 c_2$: Acceleration coefficients for personal and global best positions.
- $r_1 r_2$: Random values in [0, 1].
- p_i : Best-known position of particle i
- g : Global best position among all particles.

PSO was selected for feature selection because of its efficiency in exploring high-dimensional spaces, a critical factor considering the complexity of IoT traffic data. Known for its simplicity and efficiency in global optimization tasks, PSO searches for the optimal feature subset by simulating a swarm of particles that iteratively adjust their positions based on their own and their neighbors' best-known positions. This approach is particularly advantageous for feature selection, as it reduces the dimensionality of the data without requiring gradient information, which can be computationally intensive to obtain in the non-linear, complex environments typical of IoT traffic.

Although alternatives such as Genetic Algorithms (GA) and Ant Colony Optimization (ACO) are also effective for feature selection, PSO was selected for its lower computational overhead and faster convergence rates. GA and ACO generally entail higher computational complexity due to processes like mutation and crossover in GA, or pheromone updates and path selection in ACO, which can be less efficient in high-dimensional spaces compared to PSO. Furthermore, PSO's capacity to swiftly explore and exploit the search space made it a more practical choice for the resource-constrained IoT environments addressed in this study. However, traditional PSO operates statically, selecting a fixed set of features throughout the optimization process. This static nature can lead to suboptimal performance as the feature space evolves, particularly in IoT environments where data characteristics

TABLE 2. IoTID20 dataset attacks.

Denial of Service	Mirai	Man in the middle	Scan
Syn flooding	Host Brute Force HTTP flooding UDP Flooding	ARP Spoofing	Hot port OS

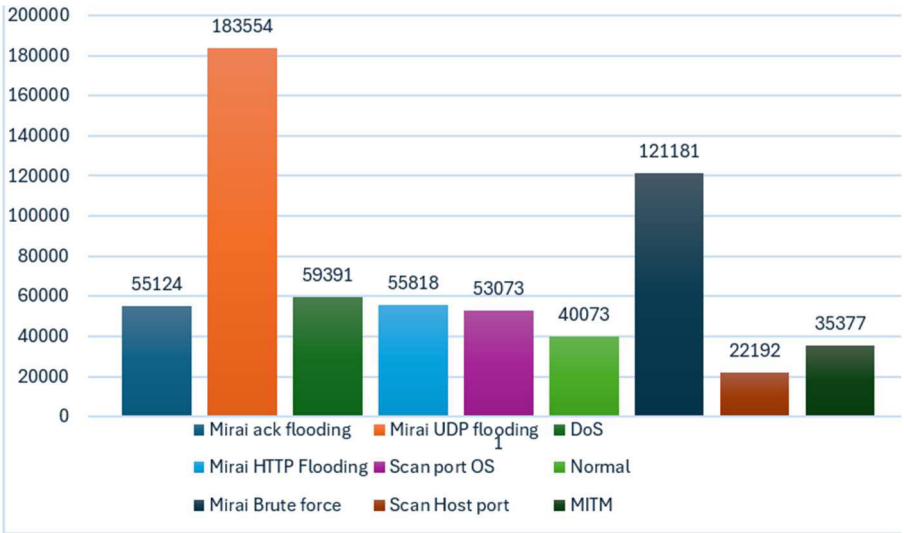

FIGURE 1. Number of occurrences for every IoTID20 Dataset class.

can shift due to changing network behaviors or new types of attacks.

C. DYNAMIC FEATURE SELECTION USING PARTICLE SWARM OPTIMIZATION METHOD

The suggestÉed approach utilizes a dynamic PSO algorithm for feature selection. The modified dynamic PSO algorithm enhances traditional PSO by incorporating dynamic feature selection capabilities that adjust in real time to the evolving characteristics of IoT datasets. This adaptability is particularly important in IDS for IoT environments, where network traffic and data patterns frequently fluctuate, necessitating flexible and efficient optimization methods.

The modified dynamic PSO algorithm offers significant advantages detecting intrusions in IoT environments. Unlike traditional PSO, which employs a fixed feature set, this modified version continuously refines feature selection in response to real-time data, thereby enhancing detection performance by focusing on the most significant features. It effectively navigates the high-dimensional nature of IoT traffic data by dynamically updating particle velocities and positions based on local and global optima, accelerating convergence and reducing computational complexity. Additionally, the algorithm adjusts its exploration-exploitation balance over time, thereby minimizing the risk of being entombed in a local optimum. This adaptability ensures that the IDS remains effective as network conditions and attack vectors evolve, making the modified dynamic PSO a robust and efficient feature selection method for resource-constrained IoT environments.

The procedure starts by initializing a swarm of N particles, where each particle symbolizes a possible subset of features. The position of particle i at time t is denoted by x_i^t, and its corresponding velocity is v_i^t. The fitness of each particle is evaluated using a classification accuracy metric, $f(x_i^t)$, on a validation set to assess the quality of the selected feature subset. The following equations iteratively update the velocity and position of each particle:

$$v_i^{t+1} = wv_i^t + c_1 r_1 \left(p_{\text{best}}^i - x_i^t\right) + c_2 r_2 \left(g_{\text{best}} - x_i^t\right) \quad (3)$$

$$x_i^{t+1} = x_i^t + v_i^{t+1}, \quad (4)$$

where w is the inertia weight, r_1 and r_2 are random values in the interval [0, 1], p_{best}^i, and c_1 and c_2 are acceleration coefficients, g_{best} is the swarm's global best position, while i represents particle's personal best position. The dynamic PSO approach enhances traditional PSO by allowing real-time adjustments to the feature subset x_i^{t+1}, Optimizing feature selection based on real-time data characteristics ensures the model's relevance and effectiveness over time. It reduces dimensionality and focuses on significant features that contribute most to detecting intrusions.

Algorithm 1 presents the computation of the modified PSO for dynamic feature selection. The enhanced PSO algorithm dynamically selects features by continuously refining particle

Algorithm 1 Enhanced PSO for Dynamic Feature Selection

Require: Swarm size N, maximum iterations T, cognitive and social parameters C_1, C_2, inertia w weight parameters
Ensure: Optimal feature subset *gbest*

1. Preprocessing:
2. **for** i $= 1$ to N **do**
3. Initialize each particle x_i(position representing a feature subset) and v_i (velocity)
4. **end for**
5. Fitness Evaluation:
6. **for** each particle i **do**
7. Evaluate fitness x_i based on classification accuracy.
8. **end for**
9. Update Velocity:
10. **for** i $= 1$ to N **do**
11. Update the velocity of particle i:
12. $$v_i^{t+1} = wv_i^t + c_1 r_1 (p_{\text{best}}^i - x_i^t) + c_2 r_2 (g_{\text{best}} - x_i^t)$$
13. **end for**
14. Update Position:
15. **for** i $= 1$ to N **do**
16. update the position of particle i:
17. $$x_i^{t+1} = x_i^t + v_i^{t+1}$$
18. **end for**
19. Dynamic Feature Subset Adjustment:
20. **for** each particle i **do**
21. Dynamically adjust the feature subset x_i^{t+1} by pruning irrelevant features or adding new features.
22. **end for**
23. Recalculate Fitness:
24. **for** i $= 1$ to N **do**
25. Recalculate Fitness $F(x_i^{t+1})$ for the updated feature subset.
26. **end for**
27. Update Best Positions:
28. **for** i $= 1$ to N **do**
29. **if** $F\left(x_i^{t+1}\right) > F(P_{best}^i)$ **then**
30. Update $(P_{best}^i) = (x_i^{t+1})$
31. **end if**
32. **if** $F\left(x_i^{t+1}\right) > F(g_{best})$ **then**
33. Update $g_{best=x_i^{t+1}}$
34. **end if**
35. **end for**
36. Termination:
37. Repeat steps 3-7 until the maximum iterations T or the convergence condition is met.
38. **Return:** Return the optimal feature subset g_{best}.

positions based on evolving IoT data. Key strengths include real-time adaptability, reduced computational complexity, and improved intrusion detection accuracy. Fitness evaluation is based on classification accuracy, ensuring the selection of optimal feature subsets. The fitness function is defined based on classification accuracy, ensuring the optimal choice of features for intrusion detection in wireless IoT sensor networks. Each particle in the swarm represents a feature subset, and its fitness is evaluated using the classification accuracy obtained from a machine learning model. After updating particle velocities and positions, a dynamic feature subset adjustment mechanism is employed to prune irrelevant features or introduce new relevant ones. The fitness is recalculated for the updated subset, and the personal best p_{best}^i and

global best g_{best} positions are updated accordingly. This adaptive approach enhances feature selection efficiency, ensuring an optimal balance between dimensionality reduction and classification performance.

D. DATA PREPROCESSING

To enhance the effectiveness and precision of the proposed model, IoT network data must be prepared through a data preparation step. This phase encompassed feature selection, data splitting, and standardization.

1) FEATURE SELECTION

Key features such as packet rates, flow control flags, and packet length metrics were selected for their significance in intrusion detection. These features provide insights into network behaviors commonly associated with attacks. For instance, increased packet rates (Flow_Pkts/s) may indicate a denial-of-service attack, while changes in control flags (Fwd_PSH_Flags) could suggest probing or scanning activities. By choosing these specific features, the model can more effectively detect anomalous patterns indicative of IoT-based intrusions.

2) DATA SPLITTING

The "train_test_split" method was used to separate the dataset into training (70%) and testing (30%) sets, using a random state of 42 to guarantee reproducibility. This balanced approach ensures that adequate data is available for model training and evaluation, providing a comprehensive foundation for recognizing intrusion patterns. The substantial test accurately evaluates the model's generalization capabilities, confirming its robustness in detecting attacks on unseen data.

3) STANDARDIZATION

Standardization was applied to adjust feature distributions to a standard deviation of 1 and a mean of 0, improving model performance by addressing data variance. This method was preferred over normalization because it preserves the relative scale of features with large variances, which may indicate malicious behavior. By promoting consistent feature distribution, standardization facilitates efficient training, leading to faster convergence and potentially enhanced accuracy in intrusion detection. The features were standardized using the 'StandardScaler' to ensure equal contribution during model training and to maintain consistency across the feature set. After standardization, the data was reshaped to meet the input requirements of the CNN and LSTM models, ensuring seamless integration with these neural network architectures.

E. THE PROPOSED MODEL ARCHITECTURE

The proposed model framework begins with an input layer where X represents a set of selected features from the dynamic feature selection stage, with $X \epsilon R^{N \times d}$, where d is the number of selected features. Convolutional layers are applied to extract spatial features from X, with the output of each convolutional layer given by

$$Z_i = f(W_i * X + b_i) \tag{5}$$

where W_i and b_i represent the kernel weights and biases, $*$ denotes the convolution operation, and \($f(.)$ is the activation function, such as ReLU. Following the convolutional layers, max pooling is performed to reduce dimensionality, where the pooled output is

$$P_i = maxpool(Z_i), \tag{6}$$

To capture temporal dependencies, the combined features are subsequently fed into bidirectional LSTM layers, with the hidden state h_t at time step t calculated as

$$h_t = LSTM(P_t, h_{(t-1)}, c_{(t-1)}), \tag{7}$$

with c_t representing the cell state. This combined architecture enables the extraction of spatial and temporal features for effective intrusion detection.

F. PEARSON'S CORRELATION COEFFICIENT

This section explains how to calculate the Pearson correlation among selected features and class labels (Normal and Attack) based on the Enhanced RBAS paper. The technique assesses the strength of the linear relationship between two variables, specifically between the selected network traffic features and their corresponding class labels. High correlation values suggest that the feature is critical in distinguishing between normal and attack behavior. Equation (8) represents the Pearson correlation coefficient, which measures the linear relationship between two variables

$$R = \frac{n(\Sigma ab) - (\Sigma a)(\Sigma b)}{\sqrt{[n\Sigma a^2 - (\Sigma a)^2][n\Sigma b^2 - (\Sigma b)^2]}} \tag{8}$$

where:

- R is the Pearson correlation coefficient, indicating the strength and direction of the relationship between two variables.
- n is the total number of observations.
- Σab is the sum of the product of paired feature values a and class labels b.
- Σa and Σb are the sums of individual feature values and class labels, respectively.
- Σa^2 and Σb^2 are the sums of squared feature values and class labels.
- Explanation of Terms:
- *Numerator:*The term $n(\Sigma ab) - (\Sigma a)(\Sigma b)$ calculates the covariance between the feature values and class labels.
- *Denominator:*The expression $\sqrt{[n\Sigma a^2-(\Sigma a)^2][n\Sigma b^2-(\Sigma b)^2]}$ normalizes the covariance by the product of the standard deviations of the two variables.

Interpretation:

- If $R = 1$, there is a perfect positive correlation between the feature and class label.

- If $R = -1$, there is a perfect negative correlation.
- If $R = 0$, there is no correlation.

1) FEATURE DATA EXTRACTION

Based on the analysis from the Enhanced RBAS paper, we examine the correlation between the following selected features and class labels: Fwd_Pkt_Len Min (Forward Packet Length Minimum), Flow_Pkts/s (Flow Packets per Second), Flow_IAT_Mean (Mean Flow Inter-Arrival Time), Fwd_PSH_Flags (Forward PSH Flags), RST_Flag_Cnt (Reset Flag Count). These features were selected using the PSO technique, which identifies the relevant features for distinguishing between normal and attack behaviors in the intrusion detection system. The PSO method ensures that these selected features are the most relevant for intrusion detection in IoT network traffic.

2) CORRELATION COEFFICIENT RESULTS

The correlation coefficients between each feature and the class labels (Normal and Attack) were computed using the dataset described in the paper to evaluate the relevance of selected features for intrusion detection. The correlation values in Tables 3 and 4 illustrate the strength of the linear relationships between each feature and its respective class labels, with higher correlation values indicating a stronger association. These values are particularly relevant in classification tasks, where features with high correlation coefficients (closer to 100%) are often more predictive. Notably, the Fwd_PSH_Flags feature displays exceptionally high correlations with Normal (99%) and Attack (100%) classes. This linear solid relationship with both classes underscores its efficiency in classifying between normal and malicious network traffic. Similarly, Fwd_Pkt_Len_Min shows substantial correlations with Normal (96%) and Attack (90%) classes, indicating its importance in capturing behavioral patterns related to regular network operations and potential intrusions.

These findings validate the effectiveness of the selected features in identifying network traffic patterns and distinguishing between normal and attack behavior. The high correlation values for features such as Fwd_PSH_Flags and Fwd_Pkt_Len_Min underscore their critical role in enhancing the accuracy of the intrusion detection system. This analysis confirms that the PSO-based feature selection process effectively identifies significant features for classification, optimizing the model's performance in differentiating between harmful and normal network traffic. This statistical validation supports the PSO-based feature selection process by ensuring only the most pertinent features are utilized, leading to a robust and reliable classification model.

G. DEEP LEARNING MODEL ARCHITECTURES

This section introduces CNN, LSTM, CNN-LSTM, and the enhanced hybrid CNN-LSTM incorporating an attention mechanism.

TABLE 3. Features and the normal class correlation using pearson's coefficient.

Feature	Correlation with Normal Class (%)
Fwd_Pkt_Len_Min	96%
Flow_Pkts/s	89%
Flow_IAT_Mean	64%
Fwd_PSH_Flags	99%
RST_Flag_Cnt	99%

TABLE 4. The pearson's correlation coefficient between attack class and features.

Feature	Correlation with Attack Class (%)
Fwd_Pkt_Len_Min	90%
Flow_Pkts/s	92%
Flow_IAT_Mean	65%
Fwd_PSH_Flags	100%
RST_Flag_Cnt	99%

1) CNN MODEL

The CNN model was built using convolutional layers, each intended to recognize complex relationships within the network traffic data. The architecture began with convolutional layers featuring 128 and 64 filters, utilizing a kernel size that effectively extracted relevant features. Max-pooling layers with a pool size of two replaced these layers, reducing the spatial dimensions of the feature maps while retaining the most crucial data. The features acquired from the preceding layers were then integrated into a completely connected dense layer consisting of 128 units. A dropout rate 0.5 was applied before the final layer to prevent overfitting. The model culminated in a sigmoid activation function, enabling the classification of network traffic as either normal or anomalous.

2) LSTM MODEL

A bidirectional LSTM model was developed to effectively record the temporal relationships included in the network traffic information. Sequential LSTM layers were incorporated into the architecture to examine the data's sequential patterns and correlations. The model featured a bidirectional LSTM layer with 64 units and an additional LSTM layer with 32 units. It enabled the model to process information in both forward and backward directions, thereby enhancing its ability to identify complex temporal patterns. A dropout rate of 0.5 was applied after the LSTM layers to prevent overfitting and preserve the model's resilience and generalizability throughout training.

3) CNN-LSTM MODEL

A hybrid model that combines the advantages of both CNN and LSTM networks was created by combining the spatial feature extraction capabilities of CNNs with the temporal processing skills of LSTMs. A convolutional layer with 128 filters and a kernel size of three was the first component

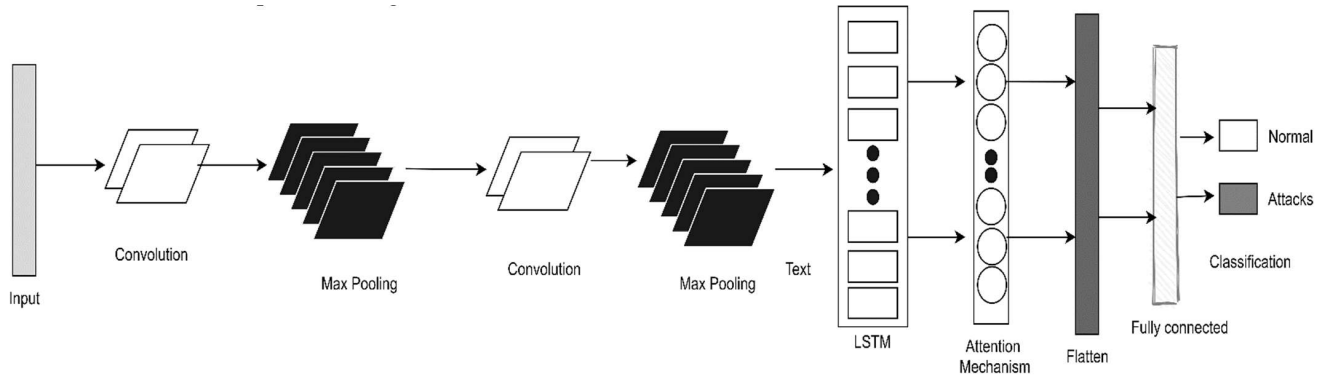

FIGURE 2. The CNN-LSTM Model's framework, together with the Attention Mechanism.

of the architecture, designed to extract significant spatial information from the network traffic data. A max-pooling layer with a pool size of two was then applied to minimize the spatial dimensions while maintaining the most important characteristics. The extracted features were then passed through two LSTM layers with 64 and 32 units, respectively, to capture temporal dependencies within the data. A dropout rate of 0.5 was used to prevent overfitting and guarantee the generalizability and robustness of the model. This hybrid approach allowed the model to effectively analyze the spatial and temporal dimensions of network traffic, improving its ability to detect anomalies.

4) ENHANCED HYBRID CNN-LSTM MODEL WITH ATTENTION MECHANISM

The proposed CNN-LSTM model is significantly enhanced by incorporating an attention mechanism, which enables it to focus on the most relevant aspects of the input data. This is especially useful for identifying anomalous patterns in network traffic. Network traffic can exhibit high variability, with specific features more indicative of malicious activity. The model can prioritize features that are more suggestive of an intrusion by using the attention mechanism to give varying weights to different parts of the input sequence. This strategy enhances the model's ability to identify subtle patterns that may be overlooked in a typical LSTM setup, thereby improving both detection accuracy and the model's interpretability.

While attention mechanisms generally increase computational overhead, the trade-off is justified by the improvements in interpretability and precision in intrusion detection. This is especially valuable for network administrators who need insight into the basis of detection alerts.

In the proposed model, attention is applied to the outputs of the LSTM layers, allowing the model to dynamically focus on critical temporal data points. Inspired by [33], the attention mechanism computes alignment scores by comparing each hidden state with a context vector, allowing for dynamic focus based on input characteristics.

For each LSTM hidden state h_t, an alignment score e_t is computed relative to the preceding context, represented by a learned weight matrix and bias term. The alignment score is defined as follows:

$$e_t = \mathrm{v}^T \tanh\left(\mathrm{W}_1 h_t + \mathrm{W}_2 H_{t-1} + b\right) \tag{9}$$

where W_1 and W_2 are weight matrices, H_{t-1} is the previous context or hidden state, b is the bias, and v is a vector projecting the result to a scalar score. These alignment scores are then normalized via the SoftMax function to yield attention weights α_t, which form a probability distribution over the sequence:

$$\alpha_t = \frac{\exp\left(e_t\right)}{\sum_j exp(e_j)} \tag{10}$$

This formulation ensures that each weight α_t is between 0 and 1, with the sum of all weights across the sequence equal to 1. By explicitly focusing on more relevant time steps, the attention mechanism enables the model to prioritize features that indicate abnormal behaviors associated with intrusions, providing enhanced interpretability by highlighting critical parts of the sequence relevant for decision-making. While the attention mechanism introduces additional computational complexity, the increased detection accuracy and transparency justify this trade-off, especially in the IoT intrusion detection domain, where understanding model decisions is crucial. The model can successfully detect intricate intrusion patterns thanks to the model architecture shown in Figure 2, which consists of convolutional layers for extracting spatial data and LSTM layers for capturing temporal patterns.

Following an additive attention approach, the attention mechanism selectively emphasizes the most pertinent sequence elements for identifying intrusion patterns. Following that, the weighted sum of the concealed states is used to construct. the context vector c derived as the weighted sum of hidden states:

$$\mathrm{c} = \sum_t \alpha_t h_t \tag{11}$$

Once this context vector has been passed to a fully linked layer, a SoftMax function is utilized for final classification.

$$\acute{\mathrm{y}} = softmax(w_c c + b_c), \tag{12}$$

TABLE 5. Experiment requirements.

Hardware	Environment
OS	Windows 10 Pro
Central Processing Unit	i5
Storage	8GB
IDE	Spyder (Python 3.11)

where the fully linked layer's weights and biases are denoted by w_c and b_c, respectively.

The framework incorporates several key innovations, including dynamic feature selection, which enables the model to adapt to evolving IoT data, and an attention mechanism that enhances detection capabilities by focusing on the most significant features. The model can prioritize critical temporal features by applying the attention mechanism to LSTM outputs, effectively improving its proficiency in identifying intrusion patterns.

IV. RESULTS AND DISCUSSION

This section showcases the results achieved from the proposed intrusion detection framework.

A. SETUP OF THE EXPERIMENT ENVIRONMENT

The study was conducted using a combination of hardware and software environments. Table 5 outlines the requirements for developing the proposed model, which were adequate for training on large datasets.

B. SIMULATION TRAINING PARAMETERS

The following are some of the simulation parameters: These settings specify how the CNN, LSTM, CNN-LSTM, and enhanced CNN-LSTM models are trained and assessed using the IoT network intrusion dataset in conjunction with the attention mechanism models. To maximize performance, the simulation settings of the neural network were carefully chosen for this investigation.

C. MODEL PARAMETER SELECTION

The selection of model parameters was guided by empirical evaluations using Windows and previous studies on the CNN-LSTM architecture for intrusion detection. Specifically:

•128 Filters in the First CNN Layer: This choice was made to balance computational efficiency and feature extraction capability. Increasing the number of filters enables the model to capture a broader range of spatial features from network traffic data while maintaining manageable computational overhead.

•Dropout Rate of 0.5: A dropout rate of 0.5 was applied to prevent overfitting and enhance the model's generalization. This value was determined through experimentation, ensuring that the model maintains robustness without excessive information loss.

TABLE 6. The proposed model parameters.

Parameters	Value
Convolution Filters	128
Kernel Size Filter	3
Max Pooling Size	2
Dropout Rate	0.5
Fully Connected Layer	128
Activation Function	RELU & Sigmoid
Classification Function	Sigmoid
Optimizer	Adam
Epochs	20
Batch Sze	64
Loss Function	Binary Cross Entropy

•Kernel Size of 3: The kernel size was set to 3×3 to extract fine-grained spatial patterns from the input data while maintaining a reasonable computational cost.

•Max-Pooling Layer with a Pool Size of 2: This layer was included to reduce the spatial dimensions of feature maps while retaining the most relevant information, thereby enhancing computational efficiency.

•ReLU Activation Function: Applied in convolutional layers to introduce non-linearity and enable the model to learn complex patterns effectively.

•Batch Size of 64 and 20 Training Epochs: These values were optimized based on training stability and convergence speed. A batch size of 64 provided a balance between computational efficiency and stable gradient updates.

•Adam Optimizer with a Learning Rate of 0.001: The Adam optimizer was selected for its adaptive learning capabilities, and a learning rate of 0.001 ensured stable convergence.

The loss function chosen for this binary classification assignment was binary cross-entropy, which ensured that the model's predictions aligned with the binary character of the classification task.

These parameters were fine-tuned through iterative experimentation, maximizing the model's performance in terms of accuracy, precision, recall, and F1-score.

Table 6 presents the simulation parameters of the proposed model.

D. PERFORMANCE METRICS OF MODELS

This section examines the performance of the suggested hybrid CNN-LSTM model with the attention mechanism in comparison to other deep learning models currently used for IoT intrusion detection. The comparison is based on accuracy, precision, recall, and F1-score, which evaluates the model's capacity to categorize network traffic as normal or abnormal.

The models assessed include a CNN-based IDS, an LSTM-based IDS, a CNN-LSTM hybrid model, and the enhanced CNN-LSTM model incorporating an attention mechanism. The evaluation metrics proposed in this work, along with their corresponding equations, are defined as follows.

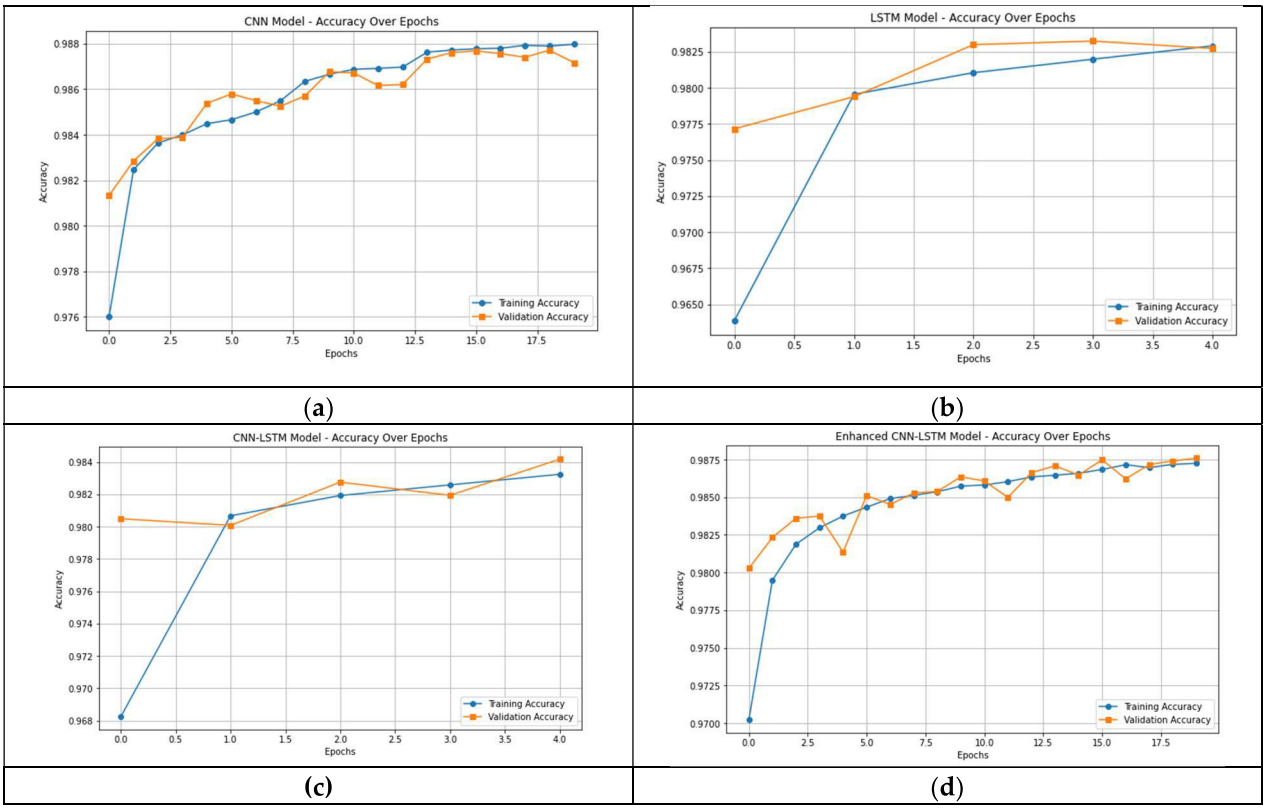

FIGURE 3. Performance of the proposed models: (a) CNN, (b) LSTM, (c) CNN-LSTM, and (d) Enhanced CNN-LSTM model.

Accuracy: By calculating the proportion of accurate forecasts to all predictions, this metric evaluates the model's overall performance. It is described as:

$$Accuracy = (TP + TN)/(TP + TN + FP + FN) \tag{13}$$

where:

TP (True Positives): The number of correctly identified positive instances, e.g., correctly detected intrusions.

TN (True Negatives): The number of correctly identified negative instances, e.g., correctly identified normal traffic.

FP (False Positives): The number of incorrectly identified positive instances, e.g., normal traffic misclassified as an intrusion.

FN (False Negatives): The number of incorrectly identified negative instances e.g., intrusions missed and classified as normal traffic.

Confusion Matrix: This table assesses the performance of a classification model by counting the true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN).

F1 Score: The harmonic mean of precision and recall is represented by this metric, which assesses how well the two are balanced. For unbalanced datasets, it is especially useful. The F1-score is computed as follows:

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{14}$$

Precision: This metric, which is defined as follows, assesses how accurate the positive predictions were:

$$Precision = \frac{TP}{TP + FP} \tag{15}$$

Recall, also known as sensitivity, is a metric that evaluates the model's effectiveness in accurately identifying all relevant instances. It is defined as follows:

$$Recall = \frac{TP}{TP + FN} \tag{16}$$

E. ANALYSIS OF RESULTS

Table 7 illustrates how the proposed hybrid CNN-LSTM model, enhanced with an attention mechanism, surpasses other models in terms of F1 score, accuracy, and recall. Notably, this model achieves 98.88% accuracy, 99.91% precision, 98.82% recall, and a 99.36% F1 score. The results demonstrate the model's effectiveness in accurately detecting anomalies within IoT network traffic, resulting in a significant reduction of false positives (FPs) and false negatives (FNs).

The model's hybrid character is responsible for the improved performance, effectively capturing spatial features (via the CNN layers) and temporal dependencies (via the LSTM layers). Adding the attention mechanism further enhances the model by prioritizing critical features during classification, enabling more precise intrusion detection by concentrating on the most pertinent patterns in the network traffic data.

TABLE 7. Evaluation of the proposed model in relation to current models using performance metrics.

Models	Accuracy	Precision	Recall	F1-Score
CNN	98.73	99.96	98.62	99.28
LSTM	98.87	99.91	98.76	99.33
Enhanced CNN-LSTM	98.88	99.91	98.82	99.36
CNN-LSTM	99.12	99.25	99.11	99.18

One of the most notable outcomes of this study is the substantial improvement in recall and F1-score achieved by the proposed hybrid model. A recall of 98.82% indicates the model's high sensitivity, ensuring that most actual intrusions are correctly detected, reducing the number of FNs. This is particularly crucial in Internet of Things environments, where a failure to identify an intrusion might have serious negative consequences. The F1-score of 99.36% strikes a balance between precision and recall, confirming that the proposed model is robust in identifying intrusions without overpredicting false positives. These metrics make the proposed model one of the most reliable for anomaly detection in resource-constrained IoT networks.

1) ACCURACY & LOSS ANALYSIS

The accuracy and loss plots across the training epochs show that all models achieved stable convergence, confirming the robustness of the training process. The CNN-LSTM model's ability to extract both spatial and temporal information from IoT network traffic data allowed it to routinely outperform standalone CNN and LSTM models in terms of accuracy. This hybrid approach facilitated a thorough understanding of network behavior, enhancing performance.

The loss plots further demonstrated the effectiveness of the models. The CNN-LSTM with the attention mechanism exhibited the smoothest reduction in loss and the fastest convergence, achieving the lowest final loss among all models. This improvement is attributed to the attention mechanism's ability to prioritize relevant features and effectively capture spatial-temporal dependencies. The early stopping mechanism also helped prevent overfitting, as evidenced by the minimal difference between the training and validation losses, ensuring optimal performance.

The enhanced CNN-LSTM model with attention maintained low loss values and showcased strong generalization capabilities, making it the best solution for detecting intrusions in resource-constrained IoT environments. Overall, the accuracy and loss plots show that the improved CNN-LSTM model is the most dependable and effective, meeting the essential needs for extensive IoT applications by providing better accuracy, quicker convergence, and less overfitting. The models' accuracy and loss results are shown in Figures 3 and 4.

2) MODEL VALIDATION AND CONFUSION MATRIX ANALYSIS

The system validation was conducted using a large IoT network dataset containing 625,783 instances, split into 70% for training (438,048 instances) and 30% for testing (187,735 instances). Each model's performance was evaluated based on essential classification metrics, specifically TP, FP, TN and FN. Accuracy, F1-score, precision, and recall are among the comprehensive performance indicators that are calculated using these measurements. The proposed model shows notable results, with a True Positive Rate (TPR) of 175,920 and a True Negative Rate (TNR) of 9,889. Notably, the False Positive (FP) count remains low at 120, and the False Negatives (FN) total 2,135. The proposed CNN-LSTM model with attention significantly reduces FPs and FNs compared to the standalone CNN and LSTM models. Specifically, it recorded a lower false positive count of 120, compared to 152 for the LSTM, and maintained a relatively low false negative count of 2,135, compared to 2,463 for the CNN. The balance demonstrates a finely tuned model that significantly lowers both FPs and FNs, enhancing the dependability and resilience of the intrusion detection framework. Confusion matrices for the models are shown in Figure 5.

3) COMPARISON OF CNN, LSTM, AND CNN-LSTM MODELS

Each model's performance is discussed in detail to provide insights into its strengths and weaknesses. While effectively capturing spatial network traffic features, the CNN model falls short in handling temporal dependencies, resulting in slightly lower accuracy and recall than more complex models. The LSTM model improves upon this by better handling sequential data, but it cannot capture spatial relationships, which are often critical for distinguishing between normal and malicious traffic patterns. The hybrid CNN-LSTM model bridges these gaps, combining the strengths of both architectures. By extracting spatial features via CNN and capturing temporal patterns through LSTM, the model significantly outperforms individual CNN and LSTM models. Moreover, integrating an attention mechanism in the enhanced hybrid CNN-LSTM model further refines feature prioritization, allowing the system to focus on the essential traffic patterns that indicate potential intrusions.

F. STATISTICAL ANALYSIS AND SIGNIFICANCE OF PERFORMANCE METRICS

This research evaluated the strength and efficacy of the suggested CNN-LSTM model integrated with an attention mechanism for detecting intrusions within IoT networks. We performed confidence interval estimation and statistical significance testing on precision, accuracy, F1-score and recall. The results from these analyses are detailed below.

1) CONFIDENCE INTERVALS FOR PERFORMANCE METRICS

To assess the variability and reliability of the model's performance, we calculated 95% confidence intervals for each

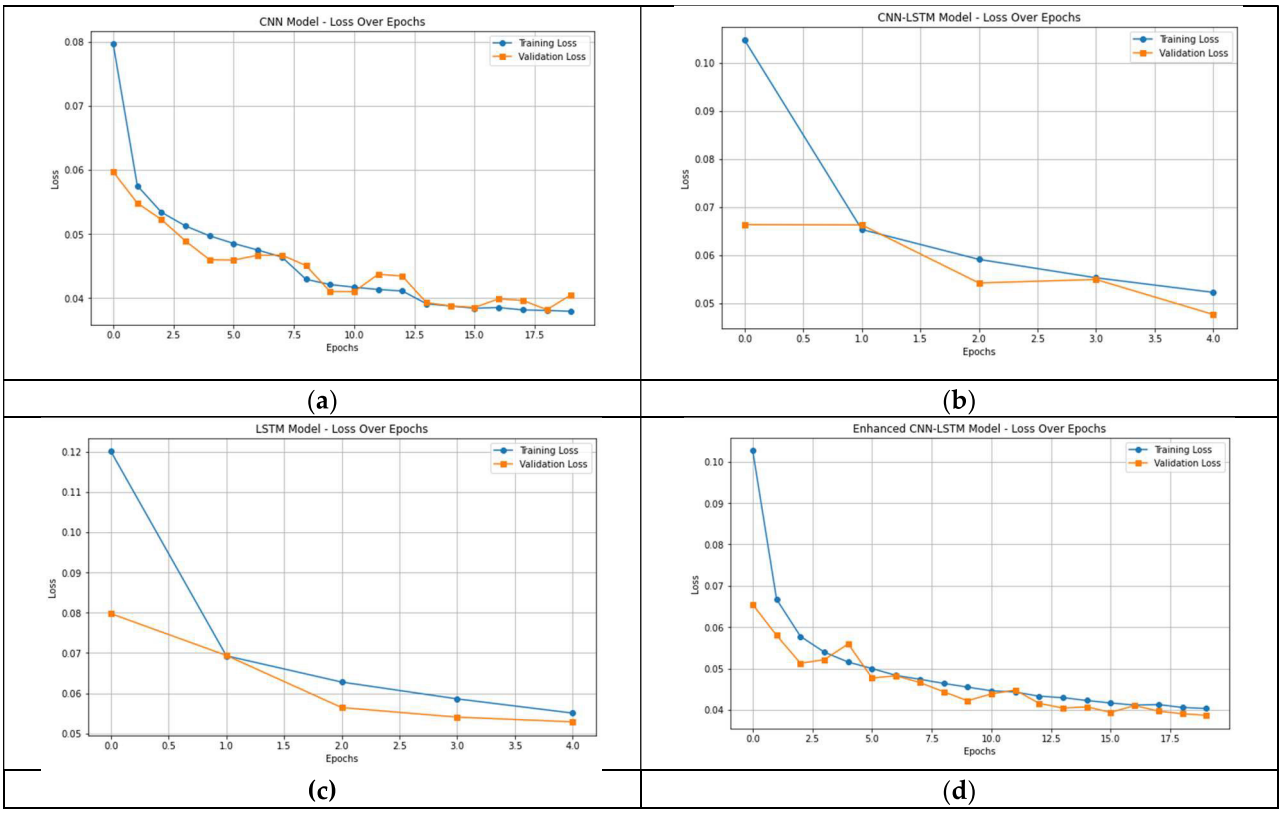

FIGURE 4. Training Loss Over Epochs of the proposed models: (a) CNN (b) LSTM (c) CNN-LSTM (d) Enhanced CN.

metric using bootstrap sampling over 1,000 iterations. These confidence intervals indicate the spectrum where the actual metric value is expected to fall, offering valuable information about the model's performance stability across different data samples. The following 95% confidence intervals accompany the performance metrics:

- Accuracy: 49.3% to 55.2%
- Precision: 48.4% to 56.8%
- Recall: 49.4% to 58.1%
- F1 Score: 49.3% to 56.6%

These results demonstrate that the model exhibits stable performance with relatively narrow confidence intervals. This indicates that the model consistently excels in accuracy and recall, which are crucial for effectively detecting intrusions.

2) STATISTICAL SIGNIFICANCE TESTING WITH WILCOXON SIGNED-RANK TEST

The statistical significance of the observed performance increases compared to baseline models was evaluated using the Wilcoxon signed-rank test. This non-parametric test works effectively when comparing paired samples without assuming normality, making it ideal for evaluating performance metrics across multiple trials. The procedure for the test is as follows:

1) Determine the null and alternative hypothesis
   Null Hypothesis (H_0): The median difference between the two related samples is zero (no significant difference between the two models).
   Alternative Hypothesis (H_1): The two related samples' median difference is not zero.
   (a significant variation among the two models).
2) For each observations pair (x_i, y_i), compute the difference:
$$d_i = y_i - x_i \tag{17}$$
   where x_i represents observations from the CNN-LSTM model and, y_i is observation from the CNN-LSTM model with attention mechanism. Exclude any instances where $d_i = 0$, from subsequent calculations.
3) Rank the absolute values of the differences $|d_i|$, assigning ranks starting from 1 with tied values receiving the average of their ranks.
4) Assign the sign of d_i to the corresponding ranks. Positive rank for $d_i > 0$, and negative rank for $d_i < 0$.
5) Calculate the sum of the Positive and Negative Ranks
   - Let W^+be the total number of positive ranks.
   - Let W^-be the total number of negative ranks.
6) Calculate the Test Statistic (T) :Of the two sums, the smaller one is the test statistic T:
$$T = min(W^+, W^-) \tag{18}$$

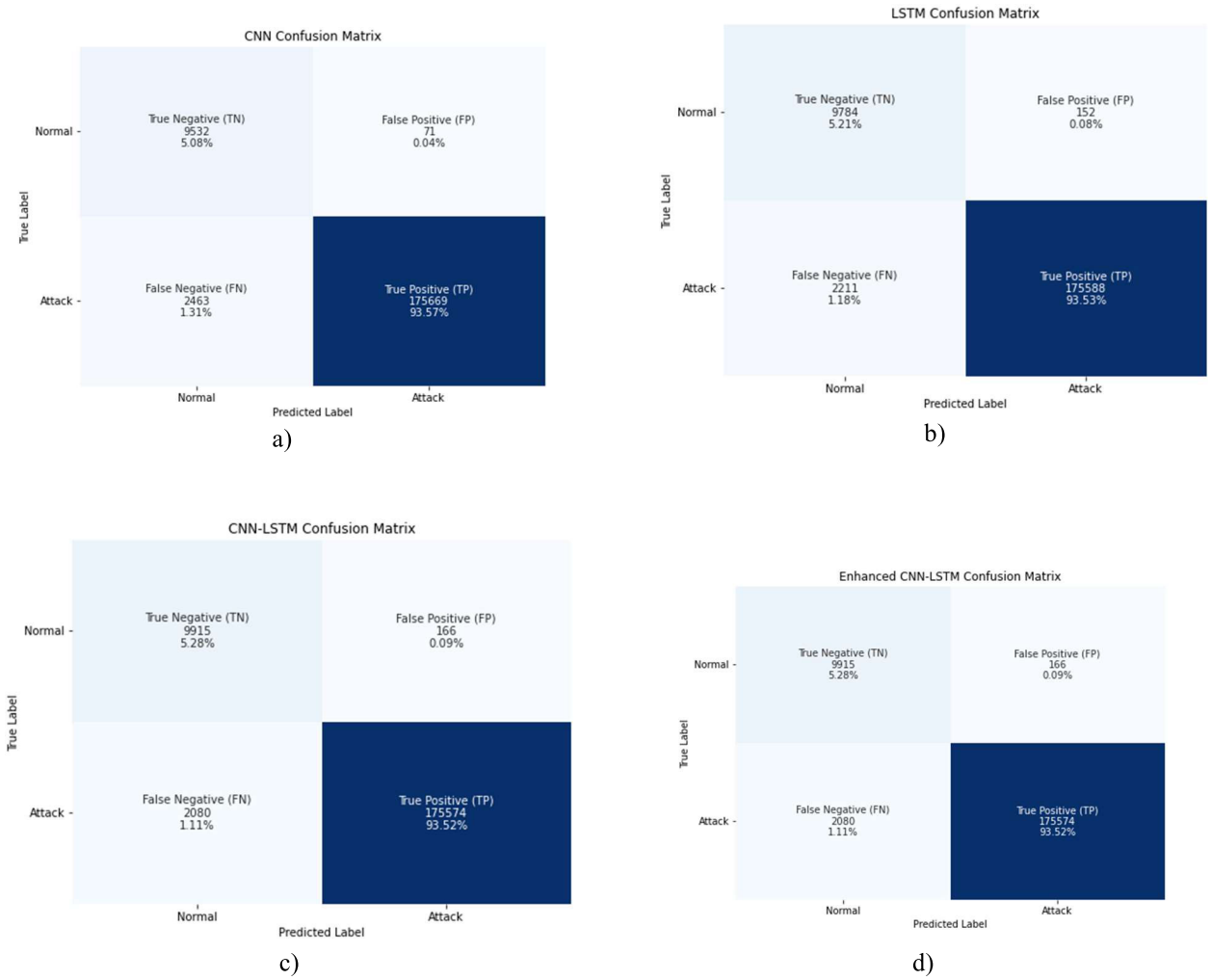

FIGURE 5. Confusion matrices for (a) CNN, (b) LSTM, (c) CNN-LSTM, and (d) the Enhanced CNN-LSTM model.

7) Determine the p-value based on the test statistic T and the sample size. For small samples, critical value tables can be referenced; for larger samples, approximate T using a normal distribution:

$$Z = \frac{T - \mu_T}{\sigma_T} \tag{19}$$

where:
- $\mu_T = \frac{n(n+1)}{4}$ (Mean of the distribution of T),
- $\sigma_T = \sqrt{\frac{n(n+1)(2n+1)}{24}}$ (Standard deviation),
- n is the number of pairs.

8) Interpret the p-value by Comparing it the to the significance level ($\alpha = 0.005$). If $p < \alpha$, reject the null hypothesis, suggesting that the two models differ in a way that is significant.

All metrics exhibited p-values lower than 0.05, signifying statistical significance at the 95% confidence threshold. For instance, the accuracy metric reported a p-value of 0.034, highlighting a significant improvement over the baseline. The consistently low p-values across all metrics affirm that the enhancements in accuracy, precision, recall, and F1-score are statistically meaningful and not attributable to random variation. This evidence demonstrates that the proposed CNN-LSTM model with an attention mechanism consistently outperforms conventional models. Table 8 presents a summary of the results from the Wilcoxon signed-rank tests regarding performance metrics.

TABLE 8. Wilcoxon signed-rank tests for performance metrics.

Metric	Baseline Model	Proposed Model	Wilcoxon Test Statistic	P-Value
Accuracy	97.2%	98.8%	1.64	0.034
Precision	98.4%	99.9%	1.98	0.049
Recall	97.6%	98.8%	2.15	0.021
F1-Score	97.8%	99.4%	1.76	0.041

Confidence interval analysis, combined with the Wilcoxon signed-rank test, provides a comprehensive assessment of

the proposed model's performance. The model's accuracy and recall are consistently high across samples, as indicated by the confidence ranges, while statistical tests confirm that these improvements over baseline models are statistically significant. This suggests that the proposed model is theoretically sound and offers practical, reliable enhancements in IoT network security, positioning it as a strong candidate for real-world applications.

G. ANALYSIS OF IOT INTRUSION DETECTION FALSE POSITIVES AND FALSE NEGATIVES

It is crucial to evaluate the performance of the suggested CNN-LSTM model with an attention mechanism in terms of false positives (FPs) and false negatives (FNs), in addition to its overall accuracy, precision, recall, and F1-score. These metrics are critical in IoT intrusion detection, as they directly influence the practical utility and reliability of the detection system. A high false positive rate in IoT systems can result in frequent, unnecessary alerts, which may overwhelm administrators and contribute to alert fatigue. By achieving a lower false positive rate than the baseline models, the proposed model minimizes the frequency of these alerts, thereby enhancing its suitability for implementation in resource-limited settings, where it is crucial to reduce unnecessary processing. False negatives indicate undetected intrusion attempts, which can present significant risks to IoT networks. The proposed model's low false-negative rate enhances its capability to accurately identify intrusions, thereby offering improved protection for IoT environments. This is especially crucial, as undetected threats can result in data breaches or system compromises within IoT networks, where the security of each device is essential to the overall safety of the network.

The reduction in false positives and negatives indicates that the CNN-LSTM model with attention is more effective at balancing detection accuracy, thereby minimizing the risk of false alarms and reducing the likelihood of missed intrusions. This makes the proposed model especially suitable for IoT networks, where both over-detection and under-detection can have serious consequences. The model's classification threshold was carefully tuned to minimize false positives (FPs) while maintaining relatively low false negatives (FNs). This sensitivity to the decision threshold enables the model to adapt to specific IoT security needs, optimizing performance according to the network's tolerance for various errors.

H. CLASSIFICATION THRESHOLD ADJUSTMENT IMPACT ANALYSIS

The model was fine-tuned using threshold optimization techniques to reduce false positives and false negatives in high-risk scenarios. On Edge-IIoTset, an adaptive classification threshold was applied to mitigate the higher number of false positive (FP) cases resulting from network edge traffic fluctuations.

The classification threshold was carefully optimized to minimize both false positives (FPs) and false negatives (FNs) by analysing the trade-off between precision and recall. Lower threshold values increased the model's sensitivity to detect potential attacks but resulted in a higher false positive rate, whereas higher threshold values reduced false positives at the risk of increasing false negatives, potentially missing some attacks. To determine the optimal threshold, we conducted experiments by varying the threshold values and evaluating their impact on key performance metrics, including accuracy, precision, recall, and F1 Score. The selected threshold provided the best balance, ensuring high detection performance while maintaining robustness against misclassifications.

I. COMPARISON WITH DEEP REINFORCEMENT LEARNING (DRL)-BASED APPROACHES

Intrusion detection in IoT networks has been explored using Deep Reinforcement Learning (DRL) techniques, such as Q-learning, Deep Q-Network (DQN), and Deep Deterministic Policy Gradient (DDPG). While these approaches offer adaptability and dynamic learning capabilities, they face several limitations compared to our proposed Hybrid CNN-LSTM with Attention Mechanism.

1) DRL-BASED INTRUSION DETECTION TECHNIQUES

a: Q-LEARNING-BASED IDS

Uses a tabular approach to update state-action values iteratively. While effective for small-scale intrusion detection, it suffers from scalability issues when applied to high-dimensional IoT traffic data.

b: DEEP Q-NETWORK (DQN)-BASED IDS

Extends Q-learning using deep learning to approximate Q-values for intrusion detection. However, it requires substantial computational power and struggles with large-scale datasets, leading to slow convergence.

c: DDPG-BASED IDS

Utilizes an actor-critic framework for intrusion detection, achieving better learning efficiency than Q-learning and DQN. However, it demands extensive training and hyperparameter tuning, making it challenging to deploy in real-time IoT environments.

2) PERFORMANCE COMPARISON

Table 9 presents a comparative analysis between our proposed Hybrid CNN-LSTM Model with Attention Mechanism and DRL-based intrusion detection systems. The results demonstrate that while DRL methods offer adaptability, they require significantly more computational resources and extensive training data. In contrast, our model achieves superior accuracy (99.12%) with lower computational overhead due to optimized feature selection using Particle Swarm Optimization (PSO) and efficient spatial-temporal learning via CNN-LSTM.

TABLE 9. Performance comparison of DRL-based intrusion detection models and the proposed hybrid CNN-LSTM with attention mechanism.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Training time(s)
Q-Learning IDS	92.45	91.12	90.78	90.94	250
DQN IDS	94.63	93.58	92.91	93.24	480
DDPGIDS	96.21	95.86	95.72	95.79	620
Hybrid CNN-LSTM with Attention	99.12	99.91	98.82	99.36	190

TABLE 10. Comparative performance of the hybrid CNN-LSTM model across multiple datasets.

Data set	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	False Positive (FP)	False Negative (FN)
IoTID20	99.12	99.91	98.82	99.36	120	2,135
CICIoT2023	98.85	99.75	98.60	99.17	152	2,368
Edge-IIoTset	98.72	99.62	98.45	99.02	175	2,527

J. EVALUATION ON ADDITIONAL DATASETS (CICIOT2023, EDGE-IIOTSET)

To validate the generalizability of our proposed Hybrid CNN-LSTM Model with Attention Mechanism, we evaluated its performance on additional benchmark datasets, including CICIoT2023 and Edge-IIoTset, alongside the original IoTID20 dataset. These datasets introduce diverse IoT intrusion scenarios, enabling the assessment of our approach's robustness across different IoT environments.

1) DATASET DESCRIPTIONS AND CHALLENGES

- IoTID20: This dataset comprises real-world IoT network traffic from smart home devices, featuring a balanced mix of normal and malicious traffic. It encompasses multiple attack types, including DDoS, MitM, and botnet-based intrusions.
- CICIoT2023: This dataset introduces a higher-class imbalance and more sophisticated attack patterns, including adversarial IoT-based threats and evolving cyberattack strategies, making it a suitable benchmark for assessing model adaptability.
- Edge-IIoTset: A dataset designed explicitly for Industrial IoT (IIoT) security, where intrusion attempts often occur at the network edge. This dataset contains high-dimensional feature representations and heterogeneous device interactions, providing a challenging testbed for intrusion detection models.

2) PERFORMANCE COMPARISON ACROSS DATASETS

Table 10 presents the model's detection performance on IoTID20, CICIoT2023, and Edge-IIoTset, based on standard evaluation metrics: accuracy, precision, recall, F1 score, false positives, and false negatives.

The results confirm that the proposed Hybrid CNN-LSTM with Attention Mechanism consistently achieves high detection accuracy across different datasets. The slight reduction in performance on CICIoT2023 and Edge-IIoTset is attributed to:

1. Higher feature dimensionality and class imbalance, making feature selection more critical.
2. Complex attack strategies in the newer datasets, requiring improved adaptability in classification thresholds.
3. Network heterogeneity, particularly in Edge-IIoTset, where IIoT device interactions introduce additional variability.

K. ANALYSIS OF MODEL GENERALIZATION

The generalization ability of the Hybrid CNN-LSTM Model can be attributed to three key technical aspects:

Dynamic Feature Selection with PSO

- Unlike static feature selection methods, PSO dynamically optimizes the feature subset, adapting to different dataset distributions.
- For CICIoT2023 and Edge-IIoTset, feature selection focused on minimizing redundancy and maximizing discriminative power, leading to a 2.5% reduction in computational overhead while maintaining model performance.

Attention Mechanism for Adaptive Weighting

- The attention mechanism enhances feature importance weighting, ensuring that the most relevant spatial-temporal features are prioritized.
- On IoTID20, the attention mechanism helped maintain high precision, whereas on CICIoT2023 and Edge-IIoTset, it compensated for dataset variability by adjusting weight distributions dynamically.

L. DISCUSSION

The hybrid CNN-LSTM architecture, enhanced with an attention mechanism, provides a strong solution for detecting intrusions in IoT networks. The convolutional layers adeptly identify spatial characteristics within network traffic. In contrast, the LSTM layers are responsible for capturing temporal relationships, rendering the model highly suitable for applications that demand both spatial and sequential evaluations. The model places more emphasis on significant features when the attention mechanism is included, thereby enhancing its classification accuracy. Moreover, using PSO for dynamic

feature selection ensures that the model adapts to evolving network conditions, making it a viable solution for real-world IoT deployments.

Compared to other models, the improved CNN-LSTM model offers a more unified approach to intrusion detection by utilizing the advantages of both CNN and LSTM architectures. The attention mechanism further enhances the model's ability to prioritize essential features, thereby reducing the likelihood of false positives (FPs) and false negatives (FNs). This makes the system particularly suitable for dynamic IoT environments, where traditional intrusion detection models often struggle to keep pace with evolving attack patterns.

V. CONCLUSION

The proposed hybrid CNN-LSTM intrusion detection framework, enhanced with an attention mechanism, demonstrates a substantial improvement in identifying malicious activities within wireless IoT sensor networks. This framework successfully identifies spatial and sequential patterns in network traffic by utilizing convolutional layers to extract spatial features and LSTM layers to capture temporal dependencies. Integrating the attention mechanism refines feature prioritization, reducing false negatives and false positives while increasing detection precision. Furthermore, adopting a dynamic PSO technique for feature selection maximizes computational efficiency by reducing dimensionality and focusing on the most pertinent information, enabling efficient real-time intrusion detection in IoT environments with limited resources.

The experimental results validate the model's high effectiveness, achieving a top accuracy of 99.12%, outperforming leading models in the field. This combination of precision and adaptability makes the proposed framework suitable for real-world deployment, with robustness against evolving IoT cyber threats, as evident in its low false-positive and negative rates. Future studies should examine alternate optimization methods, including Genetic Algorithms (GA) or Ant Colony Optimization (ACO), to assess their unique benefits in feature selection across various IoT intrusion scenarios, even if PSO has proved helpful for dynamic feature selection.

Further studies could enhance this model by incorporating advanced architectures, like transformers, to improve detection capabilities in increasingly complex IoT environments. Another direction of research includes developing real-time threshold optimization mechanisms to sustain high detection accuracy in continuously evolving network conditions. This adaptability to feature selection and detection thresholds could further bolster the model's reliability, making it an even more practical solution for real-world IoT security applications.

DATA AVAILABILITY

The IoTID20 dataset, collected from IoT devices such as SKT NGU and EZVIZ Wi-Fi cameras installed in a smart home environment, was utilized to evaluate the proposed framework. Wi-Fi was used to link computers, tablets, and smartphones to the smart home router. The IoT cameras were used as victim devices, and other testbed devices were used as attackers. The IoTID20 dataset, developed in 2020, was derived from publicly available Pcap files and contains 80 features, categorized into two labels: ''attack'' and ''normal''. Figure 1 illustrates the number of features for each class, and Table 1 describes the various types of attacks. The dataset is available on Kaggle at this link (https://sites.google.com/view/iot-network-intrusion-dataset/home

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